

- 11-10 Find the driving point impedance seen by the voltage source in Figure P11-10 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$. Insert a follower at A and repeat.

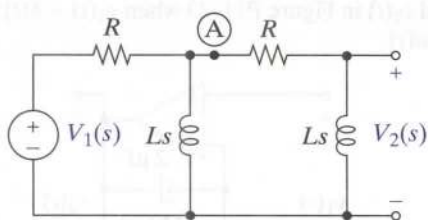


FIGURE P11-10

- 11-11 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-11. Select values of R and C so that $T_V(s)$ has a pole at $s = -100$ krad/s.

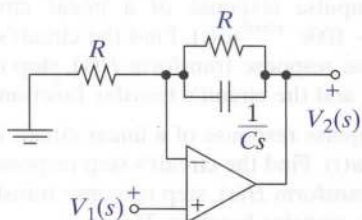


FIGURE P11-11

- 11-12 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-12. Select values of R_1 , R_2 , and C so that $T_V(s)$ has a pole at $s = -250$ krad/s and $R_2/R_1 = 100$.

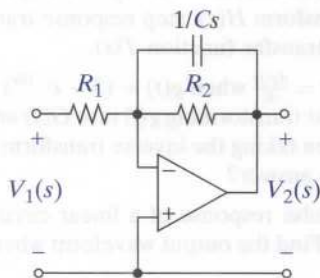


FIGURE P11-12

- 11-13 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11-13. Select values of R and L so that $T_I(s)$ has a pole at $s = -377$ rad/s.

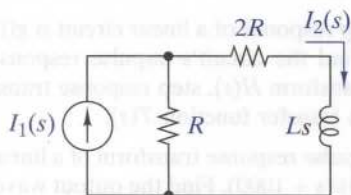


FIGURE P11-13

- 11-14 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-14. Locate the poles and zeros of the transfer function.

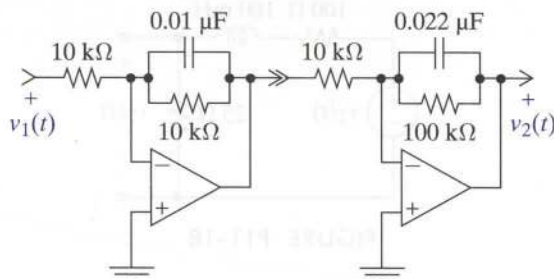


FIGURE P11-14

- 11-15 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-15. Locate the poles and zeros of the transfer function.

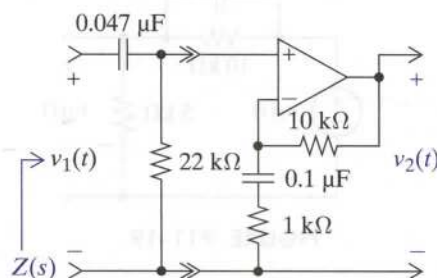


FIGURE P11-15

- 11-16 Find the input impedance $Z(s)$ in Figure P11-15.

OBJECTIVE 11-2 NETWORK FUNCTIONS, IMPULSE RESPONSE, AND STEP RESPONSES (SECTS. 11-3, 11-4)

- Given a first- or second-order linear circuit, find its impulse or step response.
- Given the impulse or step response of a linear circuit, find the network functions.
- Given the impulse or step response of a linear circuit, find the response due to other inputs.

See Examples 11-8, 11-9, 11-10, 11-11, 11-12 and Exercises 11-9, 11-10, 11-11, 11-12, 11-13, 11-14.

- 11-17 Find the impulse response at $v_2(t)$ in Figure P11-17. Find the circuit's step response.

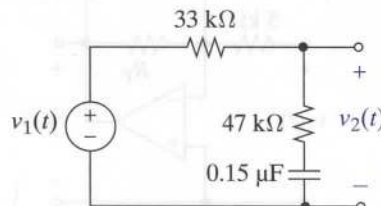


FIGURE P11-17

- 11-18 Find $v_2(t)$ in Figure P11-18 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

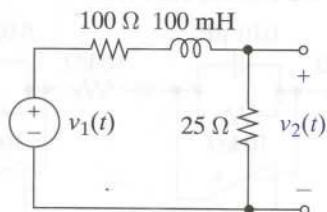


FIGURE P11-18

- 11-19 Find $v_2(t)$ in Figure P11-19 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

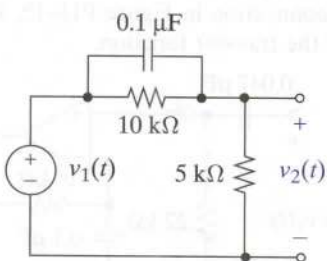


FIGURE P11-19

- 11-20 Find $h(t)$ and $g(t)$ for the circuit in Figure P11-20.

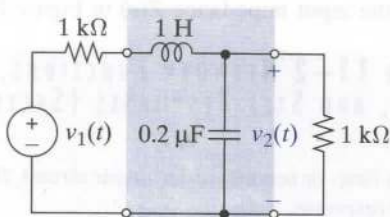


FIGURE P11-20

- 11-21 Find $h(t)$ and $g(t)$ for the circuit in Figure P11-21 if $R_F = 100 \text{ k}\Omega$.

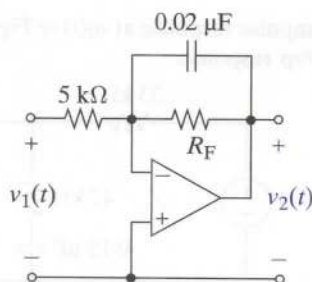


FIGURE P11-21

- 11-22 Select an appropriate R_F for the circuit of Figure P11-21 so that the step response of the circuit is $g(t) = (10e^{-1000t} - 10)u(t)$ V.

- 11-23 Find $v_2(t)$ in Figure P11-23 when $v_1(t) = \delta(t)$. Repeat for $v_1(t) = u(t)$.

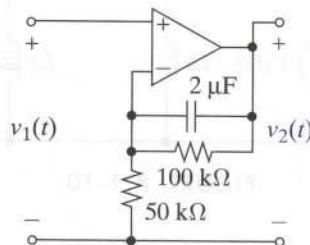


FIGURE P11-23

- 11-24 The impulse response of a linear circuit is $h(t) = (100e^{-200t} - 100e^{-1000t})\mu(t)$. Find the circuit's step response $g(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11-25 The impulse response of a linear circuit is $h(t) = \delta(t) - 2000e^{-200t}u(t)$. Find the circuit's step response $g(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11-26 The step response transform of a linear circuit is $G(s) = 1000/(s + 1000)$. Find the circuit's impulse response $h(t)$, step response $g(t)$, impulse response transform $H(s)$, and the circuit's transfer function $T(s)$.

- 11-27 The step response of a linear circuit is $g(t) = 15(e^{-20kt} - e^{-30kt})u(t)$. Find the circuit's impulse response $h(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11-28 Find $h(t) = \frac{dg(t)}{dt}$ when $g(t) = (3 - e^{-10t})u(t)$. Verify your answer by first transforming $g(t)$ into $G(s)$ and finding $H(s) = sG(s)$ and then taking the inverse transform of $H(s)$. Did you get the same answer?

- 11-29 The impulse response of a linear circuit is $h(t) = 1000[e^{-1000t}]u(t)$. Find the output waveform when the input is $x(t) = 5tu(t)$.

- 11-30 The step response of a linear circuit is $g(t) = 0.25[1 - e^{-150t}]u(t)$. Find the output waveform when the input is $v_1(t) = [20e^{-200t}]u(t)$. Use MATLAB to find the Laplace transforms of $g(t)$ and $v_1(t)$. Then find $V_2(s)$. Finally, use the inverse Laplace function to find the waveform $v_2(t)$ and plot the results.

- 11-31 The step response of a linear circuit is $g(t) = 10[e^{-50t} \cos 200t]u(t)$. Find the circuit's impulse response $h(t)$, impulse response transform $H(s)$, step response transform $G(s)$, and the circuit's transfer function $T(s)$.

- 11-32 The impulse response transform of a linear circuit is $H(s) = (s + 2000)/(s + 1000)$. Find the output waveform when the input is $x(t) = 5e^{-1000t}u(t)$. Use MATLAB to find the Laplace transform of $x(t)$. Then find $Y(s)$. Finally, use the inverse

Laplace function to find the waveform $y(f)$ and plot the results.

- 11–33 The impulse response of a linear circuit is $h(t) = 20u(t) + \delta(t)$. Find the output waveform $y(t)$ when the input is $x(t) = 2[e^{-20t}]u(t)$.

OBJECTIVE 11–3 NETWORK FUNCTIONS AND THE SINUSOIDAL STEADY-STATE RESPONSE (SECT. 11–5)

- (a) Given a first- or second-order linear circuit with a specified input sinusoid, find the sinusoidal steady-state response.
 (b) Given the network function, impulse response, or step response, find the sinusoidal steady-state response for a specified input sinusoid.

See Examples 11–13, 11–14, 11–15 and Exercises 11–15, 11–16, 11–17, 11–18.

- 11–34 The circuit in Figure P11–34 is in the steady state with $v_1(t) = 5 \cos 1414.21t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 5 \cos 1 \text{ kt}$ V. And without doing any calculations, repeat for $v_1(t) = 5$ V.

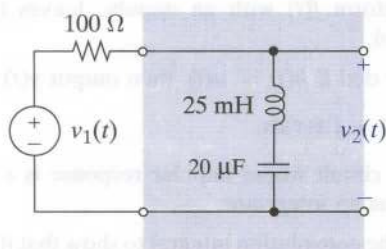


FIGURE P11–34

- 11–35 The circuit in Figure P11–35 is in the steady state with $v_1(t) = 10 \cos 500t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 10 \cos 1 \text{ kt}$ V, and for $v_1(t) = 10 \cos 10 \text{ kt}$ V. Where is the pole located?

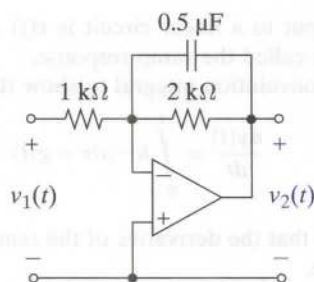


FIGURE P11–35

- 11–36 The circuit in Figure P11–36 is in the steady state with $v_1(t) = 25 \cos 2000t$ V. Find $v_{2ss}(t)$. Repeat for $v_1(t) = 25 \cos 10 \text{ kt}$ V. Where are the poles located?

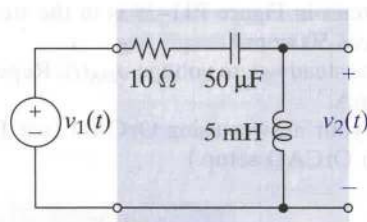


FIGURE P11–36

- 11–37 The circuit in Figure P11–37 is in the steady state with $i_1(t) = 10 \cos 50kt$ mA, $R_1 = 100 \Omega$, $R_2 = 400 \Omega$, and $L = 100 \text{ mH}$. Find $i_{2ss}(t)$. Repeat for $i_1(t) = 10 \cos 5kt$ mA. Where is the pole located?

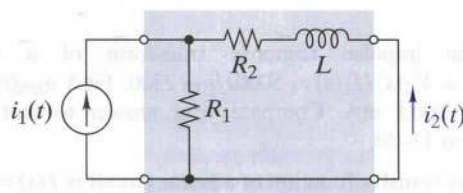


FIGURE P11–37

- 11–38 The circuit in Figure P11–38 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 0.5 \text{ }\mu\text{F}$.
 (a) Find $i_{2ss}(t)$.

(b) Verify your results using OrCAD. Use IAC for a source and note that the current comes out of the negative terminal (passive sign convention.). The IPRINT element on the right side of the figure is an ammeter and it comes from the Special Library. It has a polarity as indicated by the small “–” on the element. You should ensure that it will read the correct current direction. Before you can use the IPRINT element, it needs to be set up to display the current magnitude and phase. Double click on the element and in its property editor place a “y” in AC, MAG, and PHASE boxes. Click the Apply button and close the property editor. To obtain the simulation select AC Sweep/Noise. Set the start frequency and the stop frequency at 159.15 Hz (1000 rad/s). Place a 1 in Points/Decade. Run the simulation. Under View, select Output File. Scroll down until you see the IPRINT output.

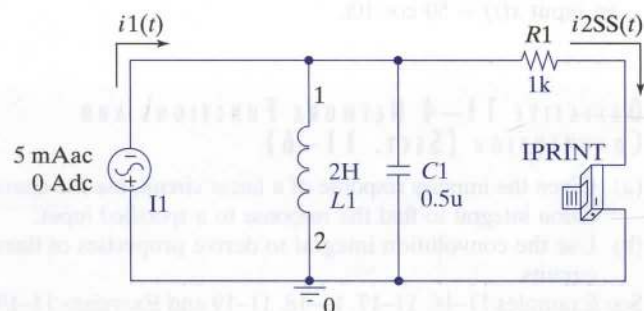


FIGURE P11–38

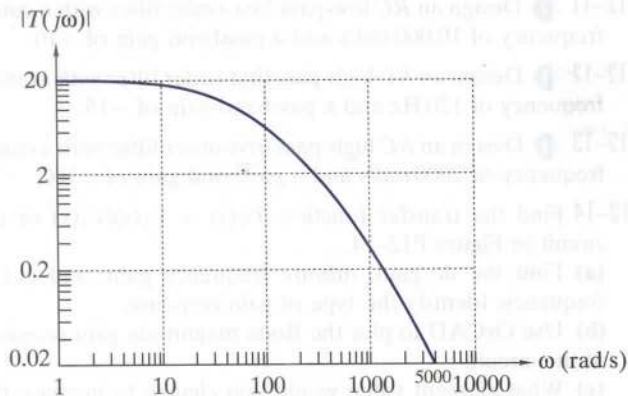


FIGURE P12-3

12-4 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-4.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximations of the gain and phase responses.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- How many dB down from the passband is the filter at one octave past the cutoff?

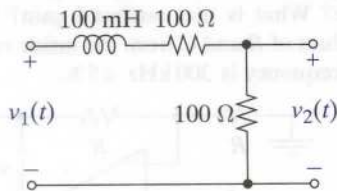


FIGURE P12-4

12-5 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-5.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- How many dB down from the passband is the filter at one octave before the cutoff?

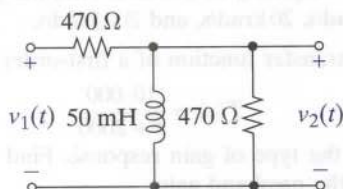


FIGURE P12-5

12-6 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-6.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.1\omega_C$, ω_C , and $10\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- What element values would you change to increase the passband gain to 10?
- How many dB down from the passband is the filter at one decade past the cutoff?

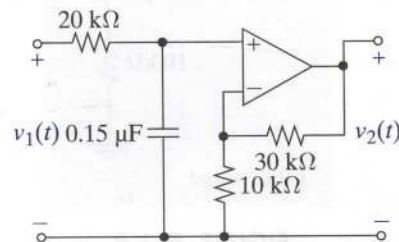


FIGURE P12-6

12-7 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-7.

- Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
- Sketch the straight-line approximation of the gain response.
- Calculate the gain at $\omega = 0.1\omega_C$, ω_C , and $10\omega_C$.
- Use OrCAD to plot the Bode magnitude gain response of the circuit. Validate your answers for part (c).
- What element values would you change to double the passband gain without changing the cutoff frequency?

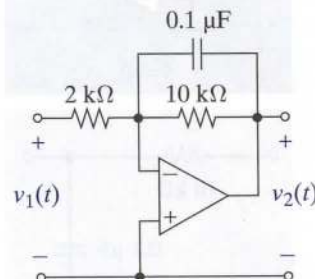


FIGURE P12-7

12-8 **E** Your task is to connect the modules in Figure P12-8 so that the gain of the transfer function is 4 and the cutoff frequency of the filter is 500 rad/s when connected between the source and the load. Repeat if the cutoff frequency is 625 rad/s and the gain did not matter.

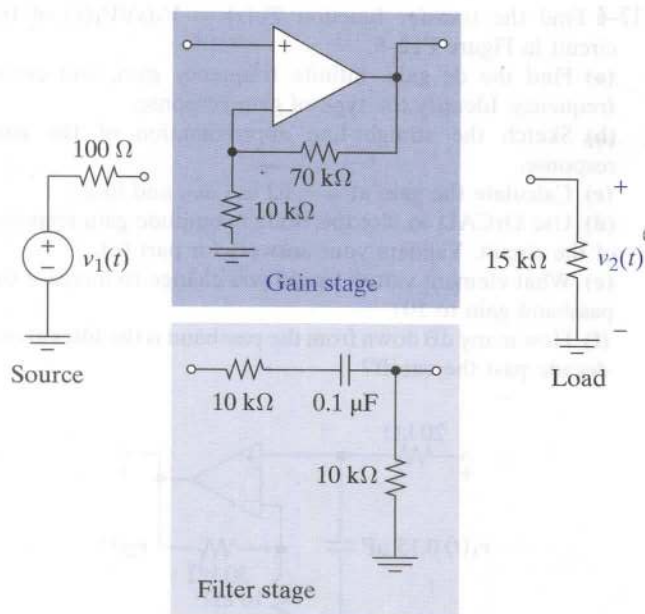


FIGURE P12-8

- 12-9 **E** A young designer needed to design a low-pass filter with a cutoff of 1 krad/s and a gain of -5 . The filter is to fit as an interface between the source and the load. The designer was perplexed when no matter how the stages shown in Figure P12-9 are connected the results are not what were expected. Explain the problem and suggest a way to achieve the desired results.

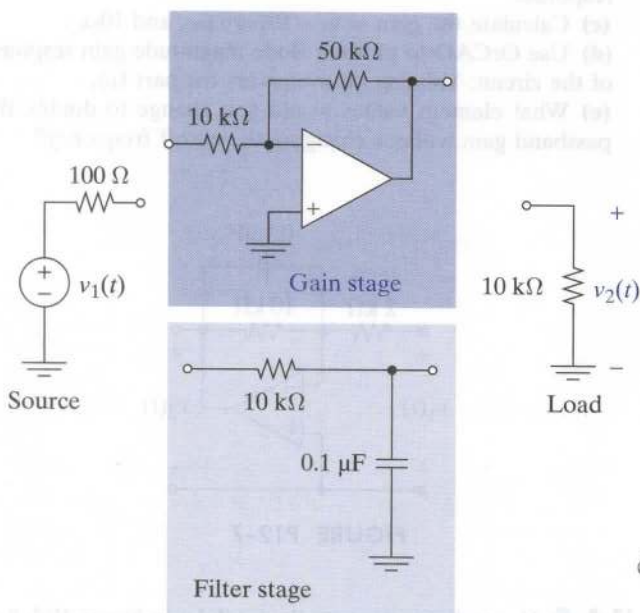


FIGURE P12-9

- 12-10 **D** Design a low-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of 1.

- 12-11 **D** Design an RC low-pass first-order filter with a cutoff frequency of 10,000 rad/s and a passband gain of $+10$.

- 12-12 **D** Design an RL high-pass first-order filter with a cutoff frequency of 120 Hz and a passband gain of -15 .

- 12-13 **D** Design an RC high-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of -100 .

- 12-14 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-14.

- (a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.
(b) Use OrCAD to plot the Bode magnitude gain response of the circuit.
(c) What element value would you change to increase the cutoff frequency by one decade?

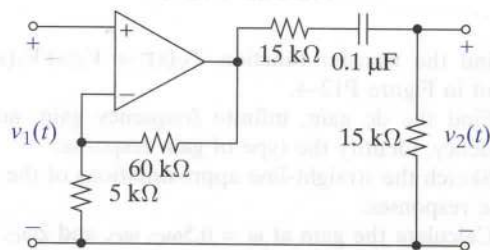


FIGURE P12-14

- 12-15 **D** Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-15. What type of gain response does the circuit have? What is the passband gain? Select practical, standard values of R and C from the inside rear cover so that the cutoff frequency is 300 kHz $\pm 5\%$.

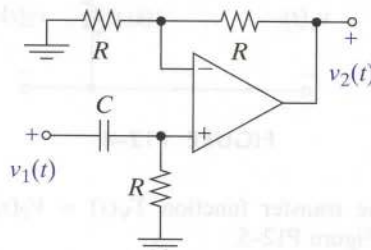


FIGURE P12-15

- 12-16 A first-order high-pass circuit has a passband gain of 20 dB and a cutoff frequency of 1000 rad/s. Find the gain (in dB) at $\omega = 0, 500, 1000$, and 5000 rad/s.

- 12-17 A first-order low-pass circuit has a passband gain of 0 dB and a cutoff frequency of 2 krad/s. Find the gain (in dB) at $\omega = 0, 200$ rad/s, 20 krad/s, and 200 krad/s.

- 12-18 **D** The transfer function of a first-order circuit is

$$T(s) = \frac{10,000}{s + 2000}$$

- (a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
(b) Use MATLAB to plot the magnitude of the Bode gain response.

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.

(b) Use MATLAB to plot the Bode magnitude of the transfer function.

(c) Design a circuit using practical values for the components that has the same transfer function.

(d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-46 For the following transfer function

$$T(s) = \frac{100(s+10)}{s(s+100)}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.

(b) Use MATLAB to plot the Bode magnitude of the transfer function.

(c) Design a circuit using practical values for the components that has the same transfer function.

(d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-47 For the following transfer function

$$T(s) = \frac{10^6 s^2}{(s+1000)^2}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.

(b) Use MATLAB to plot the Bode magnitude of the transfer function.

(c) Design a circuit using practical values for the components that has the same transfer function.

(d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-48 For the following transfer function

$$T(s) = \frac{100s(s+300)}{(s+30)(s+3000)}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency and passband gain.

(b) Use MATLAB to plot the Bode magnitude of the transfer function.

(c) Design a circuit using practical values for the components that has the same transfer function.

(d) Use OrCAD to compare the frequency response of your designed circuit with the MATLAB Bode plot.

12-49 For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T_V(s) = \frac{10(s+10)(s+100)}{(s+1)(s+1000)}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency(ies) and passband gain.

(b) Use MATLAB to plot the Bode magnitude and phase of the transfer function.

(c) What is the output $v_2(t)$, if $v_1(t) = 10 \cos(200t + 45^\circ)$ V?

12-50 For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T(s) = \frac{10^8(s+100)^2}{(s+1000)^4}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function? Estimate the cutoff frequency(ies) and passband gain.

(b) Use MATLAB to plot the Bode magnitude and phase of the transfer function.

(c) What is the output $v_2(t)$, if $v_1(t) = 10 \cos(1000t - 60^\circ)$ V?

12-51 For the following transfer function $T_V(s) = V_2(s)/V_1(s)$

$$T_V(s) = \frac{4s}{0.04s^2 + 0.2s + 1}$$

(a) Construct the straight-line Bode plot of the gain. Is this a low-pass, high-pass, bandpass, or bandstop function?

(b) Use the straight-line plot to estimate the maximum gain and the frequency at which it occurs.

(c) Use MATLAB to plot the Bode magnitude and phase of the transfer function.

12-52 Consider the gain plot in Figure P12-52.

(a) Find the transfer function corresponding to the straight-line gain plot.

(b) Use MATLAB to plot the Bode magnitude of the transfer function.

(c) Compare the straight-line gain and the actual gain at $\omega = 20$ and 200 rad/s.

(d) Design a circuit to realize the Bode plot.

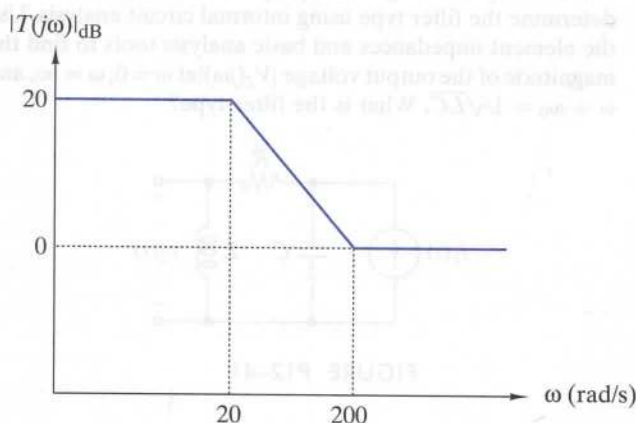


FIGURE P12-52